Electric potential in the cavity and outside

1 Problem Setup

The conductor potential V_c is already known from the conduction calculation. The same mesh also covers the air inside the loop's cavity and the air outside the conductor. These are separate regions; together call them Ω_a. The app assumes uniform permittivity ϵ and no volume charge in air. Gauss's law and $\boldsymbol{E} = -\nabla V$ give

$$\nabla \cdot (\epsilon \boldsymbol{E}) = 0 \quad \Longrightarrow \quad \nabla^2 V = 0 \quad \text{in } \Omega_a. \tag{1}$$

The distant rectangular boundary is held at $V = 0$ to approximate open space. Potential is continuous at a conductor–air surface, so its air value there must equal the conductor value.

There is one ambiguity: conduction determines V_c only up to a constant. Shifting the conductor potential relative to the fixed far boundary changes the electric field outside. Write the conductor–air boundary value as $V_c + C$. The unknown C will be chosen so the circuit has zero net charge.

2 Solving the air equation for a given boundary value

For air, the generic equation in `fem.tex` has $c = 1$ and $\boldsymbol{f} = 0$. Every air boundary has a prescribed potential: the conductor surfaces have $V_c + C$, and the far rectangle has zero. We choose test functions w as the basis functions of interior air vertices. These basis functions are zero on the boundary, so the boundary integral term in `fem.txt` vanishes. The resulting free-vertex equations are

$$\int_{\Omega_a} \nabla V \cdot \nabla w \, dA = 0. \tag{2}$$

Each air triangle contributes $K_{ij}^{(h)} = A_h \nabla N_{h,i} \cdot \nabla N_{h,j}$, with no volume-source contribution. The same assembled system contains both the cavity and outside triangles. Each air region has its own prescribed boundary values.

We do not yet know C, but this equation is linear. Solve it first for two sets of boundary values:

$$U|_{\text{conductor}} = V_c, \qquad U|_{\text{far}} = 0, \tag{3}$$

$$H|_{\text{conductor}} = 1, \qquad H|_{\text{far}} = 0. \tag{4}$$

Both U and H satisfy the zero-source air equation. The solution for any offset is therefore

$$V_a = U + CH. \tag{5}$$

Inside the enclosed cavity $H = 1$, so changing C shifts the cavity potential without changing its electric field. Outside, H changes from one at the conductor to zero at the far boundary.

3 Finding the offset from charge neutrality

The far boundary surrounds the whole circuit. By Gauss's law, the total enclosed charge per unit depth is

$$Q' = \epsilon \oint_{\Gamma_{\text{far}}} \boldsymbol{E} \cdot \boldsymbol{n}\, ds = -\epsilon \oint_{\Gamma_{\text{far}}} \nabla V \cdot \boldsymbol{n}\, ds. \tag{6}$$

We want $Q' = 0$, so we need the total outward flux of ∇V through Γ_{far}. To obtain it from the air solution, let $\mathcal{F}$ be the set of far-boundary vertices and choose

$$W_{\text{far}} = \sum_{i \in \mathcal{F}} N_i. \tag{7}$$

The basis functions sum to one along the far boundary, so $W_{\text{far}} = 1$ there. They are zero along the conductor surfaces, so $W_{\text{far}} = 0$ there. The integration-by-parts equation in `fem.tex`, with $c = 1$ and $\boldsymbol{f} = 0$, now gives

$$\int_{\Omega_a} \nabla V \cdot \nabla W_{\text{far}}\, dA = \int_{\partial\Omega_a} W_{\text{far}} \nabla V \cdot \boldsymbol{n}\, ds \tag{8}$$

$$= \oint_{\Gamma_{\text{far}}} \nabla V \cdot \boldsymbol{n}\, ds. \tag{9}$$

The cavity contributes no boundary term because $W_{\text{far}} = 0$ on its conductor boundary. Thus the total charge is $-\epsilon$ times this volume integral.

For the computed piecewise-linear potential, use the triangle expansion from `fem.tex`:

$$V_h = \sum_{k=0}^{2} V_{h,k} N_{h,k}. \tag{10}$$

Writing $W_{\text{far},h}$ for the restriction of W_{far} to triangle h, the corresponding flux is

$$\Phi(V) := \int_{\Omega_a} \nabla V \cdot \nabla W_{\text{far}}\, dA \tag{11}$$

$$= \sum_h \sum_{k=0}^{2} A_h V_{h,k} \nabla N_{h,k} \cdot \nabla W_{\text{far},h}. \tag{12}$$

This is linear in the potential values. The far-boundary basis functions were excluded as test functions when solving for those values, but we can use their sum to calculate the flux afterward.

Using equation (5), this flux is $\Phi(U + CH) = \Phi(U) + C\Phi(H)$. Set it to zero and solve the one remaining scalar equation:

$$C = -\frac{\Phi(U)}{\Phi(H)}. \tag{13}$$

The uniform factor ϵ cancels. This is why the code solves the air system twice: one solve gives the field caused by the conductor's varying potential, and the other gives the response to a uniform potential shift.

4 Completing the electric field

Add C to the conductor's potentials and assign $U + CH$ to air vertices. Conductor and air share boundary vertices, so the combined potential is continuous. For each air triangle,

$$\boldsymbol{E}_h = -\sum_{i=0}^{2} V_{h,i} \nabla N_{h,i}. \tag{14}$$

The constant shift leaves the conductor electric field unchanged. The app uses the completed field both inside and outside the circuit.

Code: `calculations/exterior.py`.